

Analyzing data on the level of cognitive processes

Exercise 3 - Set size effects

In this exercise we will implement the basic M3 model to the data sets from Oberauer (2019) “Working Memory Capacity Limits Memory for Bindings” in which participants completed a cued recall task. In two experiments, participants were asked to memorize lists. In different conditions, various numbers of words (set size 2, 4, 6, 8) were displayed within a list. At test, participants were cued with a list position and asked to select the word from a selection of words. This selection could contain apart from the correct word, other presented list words (IOP) or completely new words (NPL).

The data set for experiment 1 “exercise3_data_exp1.txt” and experiment 2 “exercise3_data_exp2.txt” are in the folder “data”. Here is an overview what the individual variables in each data set refer to:

- id: participant ID
- setsize: setsize of words within a list (2, 4, 6 or 8) as a factor
- setsize_lin: setsize of words within a list (2, 4, 6 or 8) as a linear variable
- n_IIP: total number of correct responses in each trial
- n_IOP: total number of IOP responses in each trial
- n_NPL: total number of NPL in each trial
- IIP: number of correct responses (IIP)
- IOP: number of errors when a word from the list was reported, but at the incorrect position (IOP)
- NPL: number of errors when a word *not* from the list was reported (NPL)
- total: total number of IIP, IOP, and NPL

Task 1: Experiment 1

We are interested in investigating how the set sizes affect participants memorizing the word lists in experiment 1 (“exercise3_data_exp1.txt”). Specifically, we are interested in examining how the cognitive processes, namely binding memory and item memory, change with increasing set sizes.

To this end, we will...

- ... examine and plot the proportions of correct responses in experiment 1 descriptively for the different set sizes. How do the proportions change for the different set sizes?
- ... implement the basic M3 model and plot the results (e.g., the model estimates, model predictions). How are binding memory and item memory affected by the different set sizes? Does this match your expectations?

Load libraries

```
# empty environment
rm(list = ls())

# load libraries
library("here")
library("bmm")
library("dplyr")
library("tidyr")
```

```
library("brms")
library("ggplot2")
library("tidybayes")
library("readr")
```

Load data

```
# experiment 1
mydata_exp1 <- read.table(here("data", "exercise3_data_exp1.txt"))
mydata_exp1$setsize <- as.factor(mydata_exp1$setsize)
```

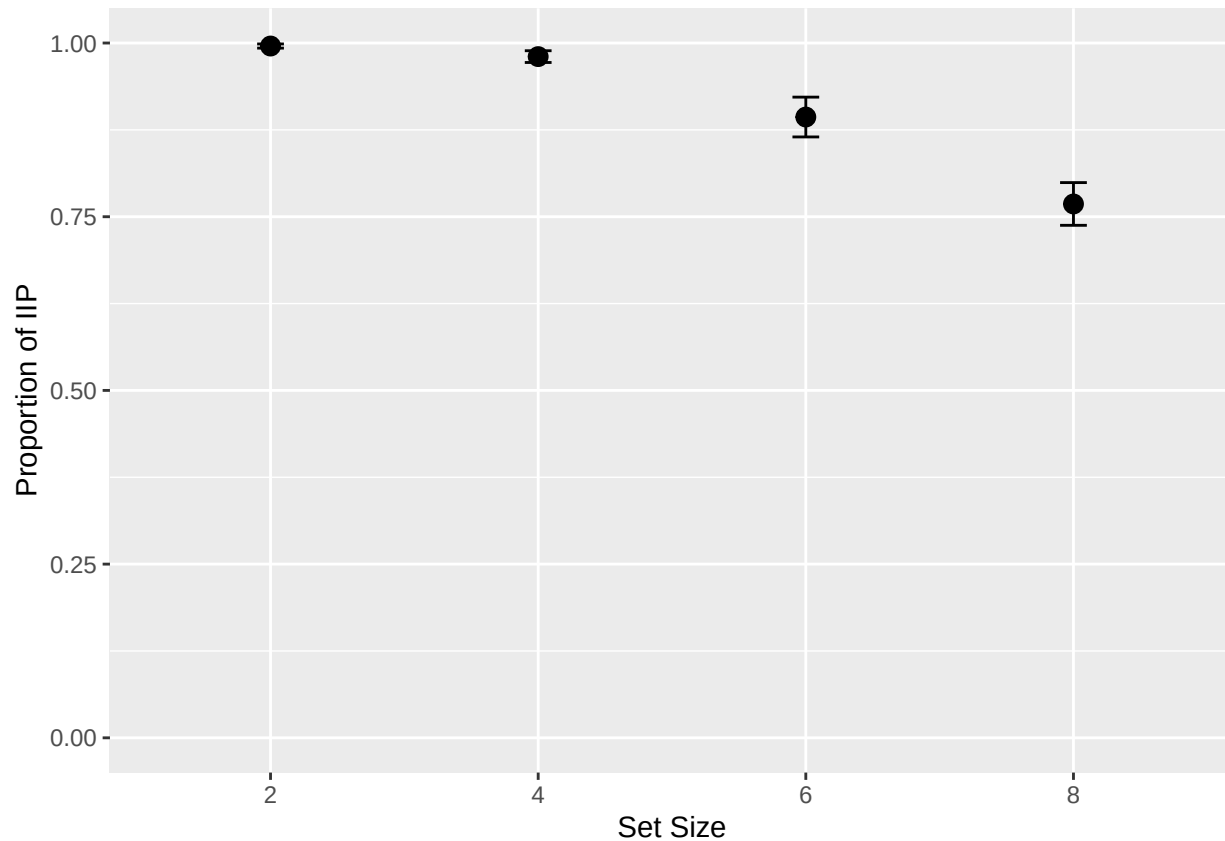
Descriptive proportions

```
# calculate proportions
prop_data <- mydata_exp1 %>%
  group_by(id, setsize) %>%
  summarise(prop = sum(IIP) / sum(total), .groups = "drop") %>%
  group_by(setsize) %>%
  summarise(
    mean_prop = mean(prop),
    se = sd(prop) / sqrt(n()),
    lower = mean_prop - 1.96 * se,
    upper = mean_prop + 1.96 * se,
    .groups = "drop"
  )
```

prop_data

```
## # A tibble: 4 x 5
##   setsize mean_prop      se lower upper
##   <fct>      <dbl>   <dbl> <dbl> <dbl>
## 1 2          0.996 0.00152 0.993 0.999
## 2 4          0.980 0.00432 0.972 0.989
## 3 6          0.893 0.0146  0.865 0.922
## 4 8          0.768 0.0157  0.738 0.799
```

```
# plot proportions
ggplot(prop_data, aes(x = factor(setsize), y = mean_prop)) +
  geom_point(size = 3) +
  geom_errorbar(aes(ymin = lower, ymax = upper), width = 0.1) +
  labs(y = "Proportion of IIP", x = "Set Size") +
  coord_cartesian(ylim = c(0, 1))
```



Implement basic M3 model

```
n_cores <- 4
n_iters <- 2000 # number of total iterations
n_warmup <- 1000 # number of warmups
n_chains <- 4 # number of chains
```

Set model settings

```
# model
m3_model_ss <- m3(resp_cats = c("IIP", "IOP", "NPL"),
  num_options = c("n_IIP", "n_IOP", "n_NPL"),
  choice_rule = "softmax",
  version = "ss")

# model formula
m3_formula_ss <- bmf(
  IIP ~ b + a + c,
  IOP ~ b + a,
  NPL ~ b,
  c ~ 1 + setsize + (1 + setsize | id),
  a ~ 1 + setsize + (1 + setsize | id)
)

# check priors
```

```
default_prior(m3_formula_ss, data = mydata_exp1, model = m3_model_ss)
```

Define model formula, priors and model

```
##           prior class      coef group resp dpar nlpar   lb   ub
##           lkj(1)  cor
##           lkj(1)  cor              id
##           normal(0,0.5)    b  setsize4          a <NA> <NA>
##           normal(0,0.5)    b  setsize6          a <NA> <NA>
##           normal(0,0.5)    b  setsize8          a <NA> <NA>
## student_t(3, 0, 2.5)    sd
## student_t(3, 0, 2.5)    sd              id          a    0
## student_t(3, 0, 2.5)    sd Intercept    id          a    0
## student_t(3, 0, 2.5)    sd  setsize4    id          a    0
## student_t(3, 0, 2.5)    sd  setsize6    id          a    0
## student_t(3, 0, 2.5)    sd  setsize8    id          a    0
##           (flat)    b
##           normal(0,2)    b  setsize4          c <NA> <NA>
##           normal(0,2)    b  setsize6          c <NA> <NA>
##           normal(0,2)    b  setsize8          c <NA> <NA>
## student_t(3, 0, 2.5)    sd
## student_t(3, 0, 2.5)    sd              id          c    0
## student_t(3, 0, 2.5)    sd Intercept    id          c    0
## student_t(3, 0, 2.5)    sd  setsize4    id          c    0
## student_t(3, 0, 2.5)    sd  setsize6    id          c    0
## student_t(3, 0, 2.5)    sd  setsize8    id          c    0
##           normal(0,2)    b
##           normal(3,1)    b Intercept          c <NA> <NA>
##           normal(0,0.5)    b
##           normal(2,1)    b Intercept          a <NA> <NA>
##           constant(0)    b Intercept          b <NA> <NA>
##           source
##           default
## (vectorized)
## (vectorized)
## (vectorized)
## (vectorized)
##           default
## (vectorized)
## (vectorized)
## (vectorized)
## (vectorized)
## (vectorized)
##           default
## (vectorized)
## (vectorized)
## (vectorized)
##           default
## (vectorized)
## (vectorized)
## (vectorized)
## (vectorized)
## (vectorized)
##           user
```

```
##      user
##      user
##      user
##      user
```

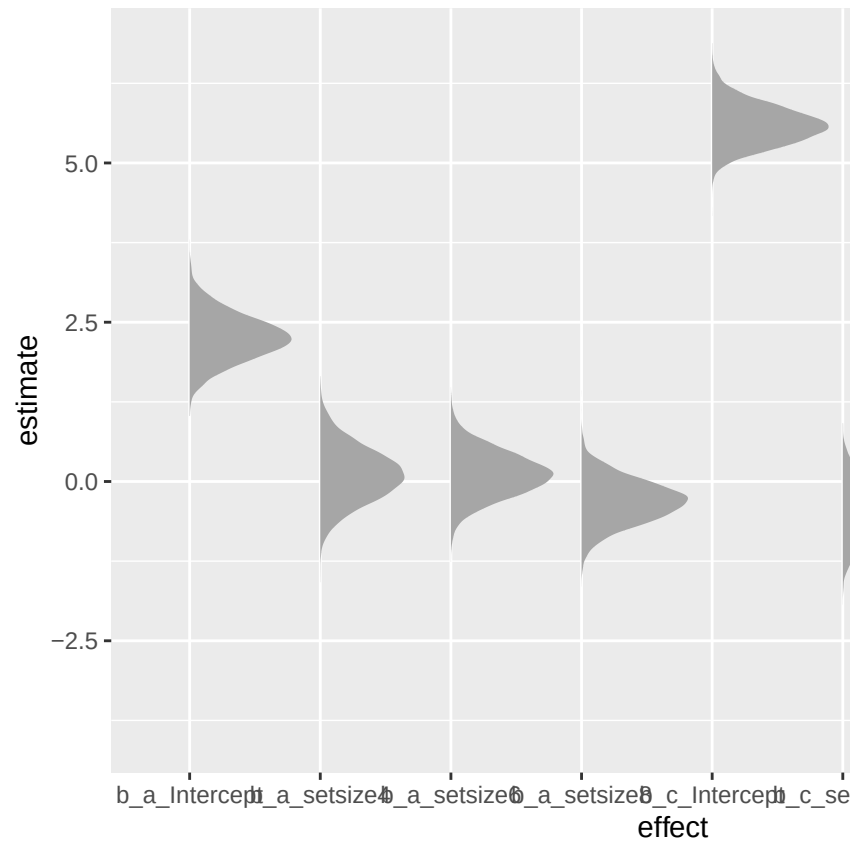
Fit and run model

```
##   Model: m3(resp_cats = c("IIP", "IOP", "NPL"),
##             num_options = c("n_IIP", "n_IOP", "n_NPL"),
##             choice_rule = "softmax",
##             version = "ss")
##   Links: c = identity; a = identity
## Formula: b = 0
##           c ~ 1 + setsize + (1 + setsize | id)
##           a ~ 1 + setsize + (1 + setsize | id)
##           IIP ~ b + a + c
##           IOP ~ b + a
##           NPL ~ b
##   Data: (Number of observations: 480)
##   Draws: 4 chains, each with iter = 3000; warmup = 1000; thin = 1;
##           total post-warmup draws = 8000
##
## Multilevel Hyperparameters:
## ~id (Number of levels: 20)
##
##           Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS
## sd(c_Intercept)      0.39      0.16      0.05      0.71 1.01      931
## sd(c_setsize4)        0.51      0.28      0.04      1.12 1.00     1777
## sd(c_setsize6)        0.44      0.19      0.12      0.86 1.00      867
## sd(c_setsize8)        0.22      0.15      0.01      0.56 1.01      615
## sd(a_Intercept)      0.67      0.33      0.06      1.36 1.00     1546
## sd(a_setsize4)        0.75      0.58      0.03      2.18 1.00     3840
## sd(a_setsize6)        0.47      0.36      0.02      1.36 1.00     2272
## sd(a_setsize8)        0.46      0.33      0.02      1.21 1.00     1431
## cor(c_Intercept,c_setsize4) 0.02      0.40     -0.72      0.78 1.00     3771
## cor(c_Intercept,c_setsize6) 0.37      0.37     -0.50      0.91 1.00     1020
## cor(c_setsize4,c_setsize6)  0.18      0.39     -0.64      0.82 1.00     1691
## cor(c_Intercept,c_setsize8) -0.11      0.44     -0.84      0.74 1.00     3304
## cor(c_setsize4,c_setsize8)  0.04      0.44     -0.77      0.81 1.00     2180
## cor(c_setsize6,c_setsize8)  0.33      0.46     -0.66      0.94 1.00      875
## cor(a_Intercept,a_setsize4) -0.04      0.44     -0.83      0.79 1.00    10775
## cor(a_Intercept,a_setsize6) -0.01      0.44     -0.80      0.81 1.00     6422
## cor(a_setsize4,a_setsize6)  0.02      0.45     -0.82      0.83 1.00     6460
## cor(a_Intercept,a_setsize8) -0.01      0.44     -0.79      0.81 1.00     5913
## cor(a_setsize4,a_setsize8)  0.04      0.45     -0.80      0.82 1.00     3685
## cor(a_setsize6,a_setsize8)  0.15      0.45     -0.76      0.88 1.00     2572
##
##           Tail_ESS
## sd(c_Intercept)      923
## sd(c_setsize4)       2544
## sd(c_setsize6)       1339
## sd(c_setsize8)       1509
## sd(a_Intercept)      1434
## sd(a_setsize4)       3108
## sd(a_setsize6)       3879
## sd(a_setsize8)       3287
```

```
## cor(c_Intercept,c_setsize4)      3714
## cor(c_Intercept,c_setsize6)      773
## cor(c_setsize4,c_setsize6)      4106
## cor(c_Intercept,c_setsize8)      3489
## cor(c_setsize4,c_setsize8)      4984
## cor(c_setsize6,c_setsize8)      1883
## cor(a_Intercept,a_setsize4)      5116
## cor(a_Intercept,a_setsize6)      5452
## cor(a_setsize4,a_setsize6)      5636
## cor(a_Intercept,a_setsize8)      5404
## cor(a_setsize4,a_setsize8)      5282
## cor(a_setsize6,a_setsize8)      5527
##
## Regression Coefficients:
##           Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
## c_Intercept      5.59      0.31      5.02      6.22 1.00      3593      4229
## c_setsize4       -0.46      0.37     -1.18      0.26 1.00      3787      5553
## c_setsize6       -2.03      0.32     -2.70     -1.43 1.00      3425      4598
## c_setsize8       -2.86      0.31     -3.49     -2.29 1.00      3504      4124
## a_Intercept      2.26      0.36      1.55      2.98 1.00      5165      5497
## a_setsize4        0.10      0.42     -0.71      0.96 1.00     10348      6081
## a_setsize6        0.11      0.35     -0.56      0.81 1.00      6661      6472
## a_setsize8       -0.31      0.34     -0.99      0.35 1.00      5753      5885
##
## Constant Parameters:
##           Value
## b_Intercept      0.00
##
## Draws were sampled using sample(hmc). For each parameter, Bulk_ESS
## and Tail_ESS are effective sample size measures, and Rhat is the potential
## scale reduction factor on split chains (at convergence, Rhat = 1).
```

```
# extract posterior samples
posterior <- as_draws_df(m3_fit_ss_exp1) %>%
  select(starts_with("b_")) %>%
  pivot_longer(cols = starts_with("b_"), names_to = "effect", values_to = "estimate") %>%
  filter(effect != "b_b_Intercept")

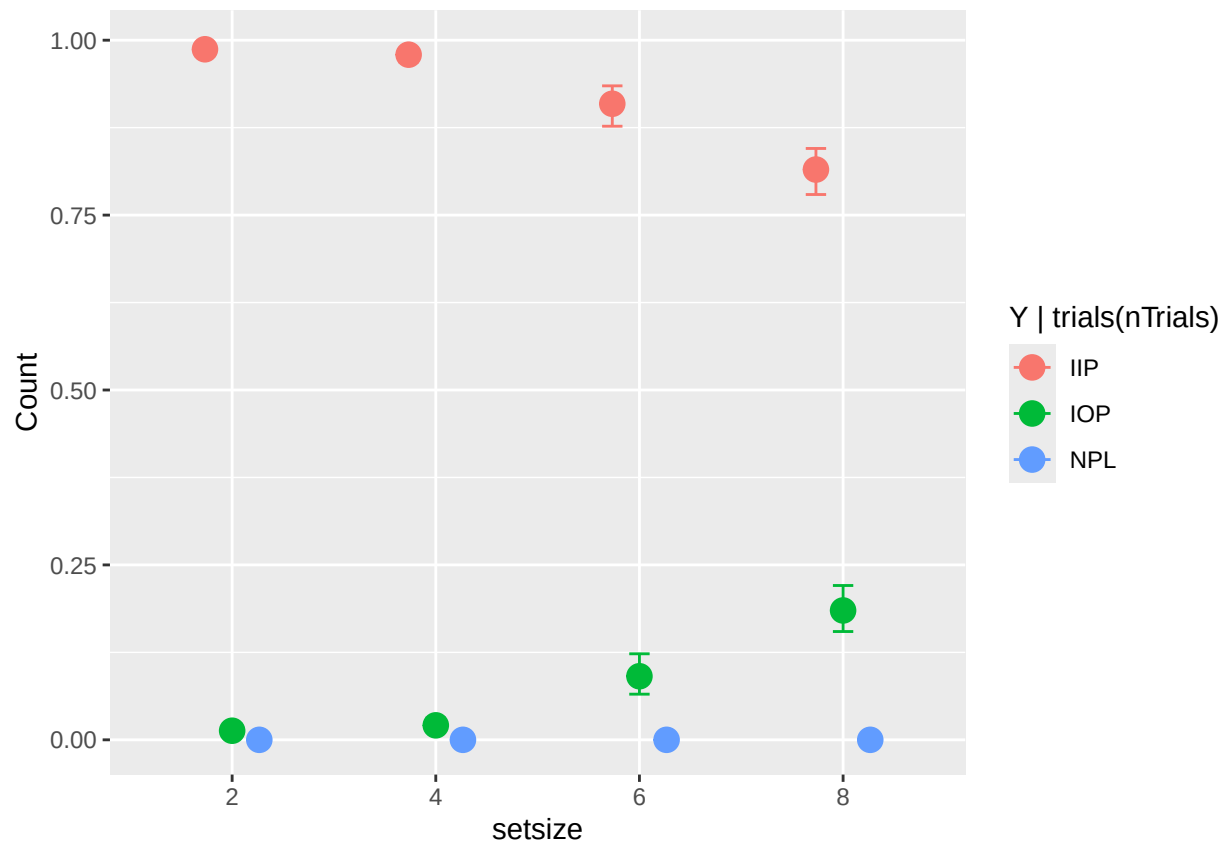
# plot posterior samples
ggplot(posterior, aes(x = effect, y = estimate)) +
  stat_slab()
```



Plot model posteriors and model predictions

plot model predictions on native scale

`conditional_effects(m3_fit_ss_exp1, categorical = T, effects = "setsize")`



```
# to-be testes hypotheses
tests <- c(
  "c_Intercept = c_setsize4",
  "c_setsize4 = c_setsize6",
  "c_setsize6 = c_setsize8",
  "a_Intercept = a_setsize4",
  "a_setsize4 = a_setsize6",
  "a_setsize6 = a_setsize8"
)

hypothesis(m3_fit_ss_exp1, tests)
```

Hypothesis tests

```
## Hypothesis Tests for class b:
##
```

	Hypothesis	Estimate	Est.Error	CI.Lower	CI.Upper	Evid.Ratio
## 1	(c_Intercept)-(c_... = 0	6.06	0.63	4.86	7.35	0.00
## 2	(c_setsize4)-(c_s... = 0	1.57	0.25	1.11	2.09	0.00
## 3	(c_setsize6)-(c_s... = 0	0.82	0.12	0.59	1.07	0.00
## 4	(a_Intercept)-(a_... = 0	2.15	0.62	0.96	3.36	0.01
## 5	(a_setsize4)-(a_s... = 0	-0.01	0.51	-0.98	1.01	1.38
## 6	(a_setsize6)-(a_s... = 0	0.43	0.33	-0.21	1.08	0.91

```
## Post.Prob Star
## 1      0.00    *
## 2      0.00    *
## 3      0.00    *
```



```
## 4      0.01    *
## 5      0.58
## 6      0.48
## ---
## 'CI': 90%-CI for one-sided and 95%-CI for two-sided hypotheses.
## '*': For one-sided hypotheses, the posterior probability exceeds 95%;
## for two-sided hypotheses, the value tested against lies outside the 95%-CI.
## Posterior probabilities of point hypotheses assume equal prior probabilities.
```

Task 2: Experiment 2

In experiment 2, the same design as in experiment 1 was applied. However, the pool from which the words in the list were chosen from was restricted in comparison to experiment. We want to examine whether and how this change from a large word pool in experiment 1 to a small word pool in experiment 2 changes the underlying cognitive processes of the set size effect.

- Thus, we will first plot the data from experiment 2 descriptively (“exercise3_data_exp2.txt”).
- Next, we implement the same basic M3 model and examine the model output. Plot the model results.
- We test whether there are differences among the different set sizes using hypothesis tests. Do the results differ between experiment 1 and 2? If yes, how so?

Load data

```
# experiment 2
mydata_exp2 <- read.table(here('data', 'exercise3_data_exp2.txt'))
mydata_exp2$setsize <- as.factor(mydata_exp2$setsize)
```

Descriptive proportions

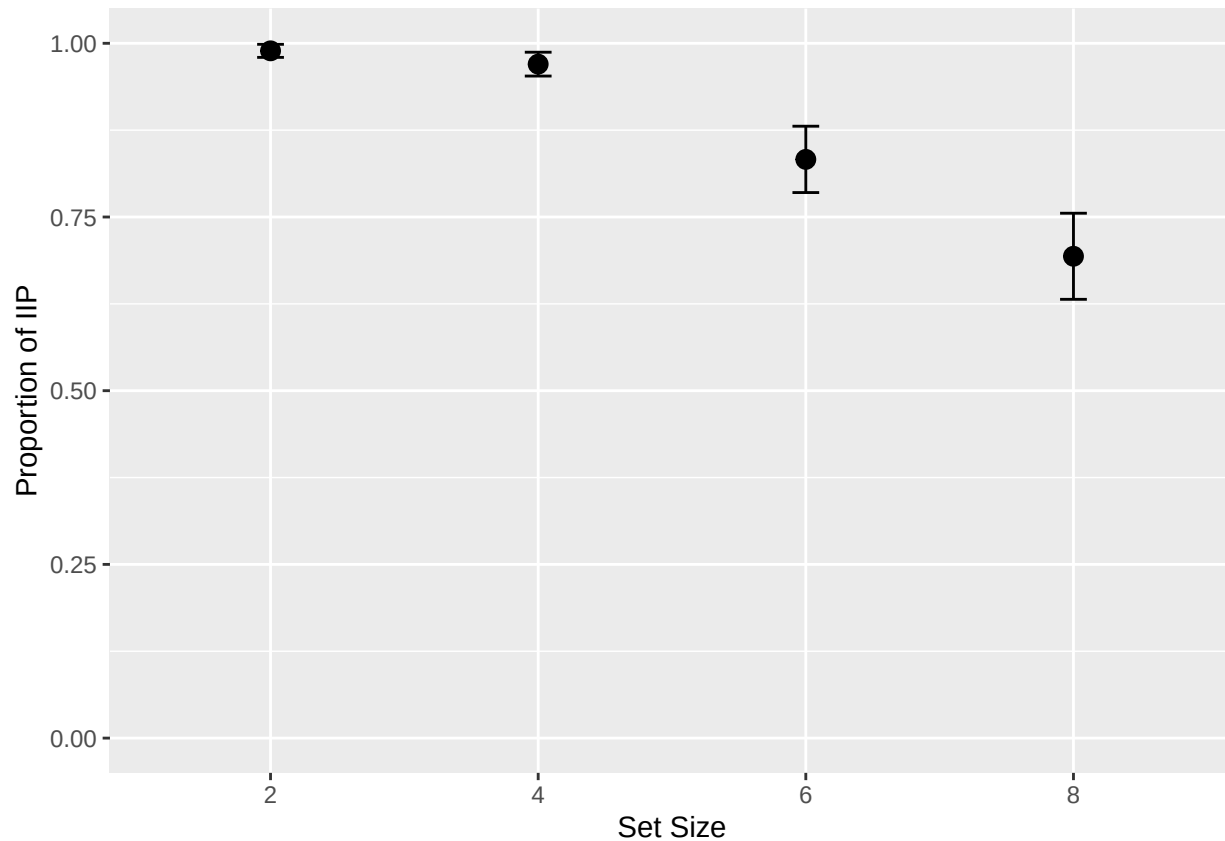
```
# calculate proportions
prop_data <- mydata_exp2 %>%
  group_by(id, setsize) %>%
  summarise(prop = sum(IIP) / sum(total), .groups = "drop") %>%
  group_by(setsize) %>%
  summarise(
    mean_prop = mean(prop),
    se = sd(prop) / sqrt(n()),
    lower = mean_prop - 1.96 * se,
    upper = mean_prop + 1.96 * se,
    .groups = "drop"
  )
```

prop_data

```
## # A tibble: 4 x 5
##   setsize mean_prop      se lower upper
##   <fct>      <dbl>    <dbl> <dbl> <dbl>
## 1 2          0.989 0.00480 0.980 0.999
## 2 4          0.97  0.00875 0.953 0.987
## 3 6          0.833 0.0244  0.785 0.881
## 4 8          0.693 0.0316  0.632 0.755
```

```
# plot proportions
ggplot(prop_data, aes(x = factor(setsize), y = mean_prop)) +
  geom_point(size = 3) +
```

```
geom_errorbar(aes(ymin = lower, ymax = upper), width = 0.1) +
labs(y = "Proportion of IIP", x = "Set Size") +
coord_cartesian(ylim = c(0, 1))
```



Implement basic M3 model

```
n_cores <- 4
n_iters <- 2000 # number of total iterations
n_warmup <- 1000 # number of warmups
n_chains <- 4 # number of chains
```

Set model settings

```
# model
m3_model_ss <- m3(resp_cats = c("IIP", "IOP", "NPL"),
  num_options = c("n_IIP", "n_IOP", "n_NPL"),
  choice_rule = "softmax",
  version = "ss")

# model formula
m3_formula_ss <- bmf(
  IIP ~ b + a + c,
  IOP ~ b + a,
  NPL ~ b,
  c ~ 1 + setsize + (1 + setsize | id),
```

```

a ~ 1 + setsize + (1 + setsize | id)
)

# check priors
default_prior(m3_formula_ss, data = mydata_exp2, model = m3_model_ss)

```

Define model formula, priors and model

```

##           prior class      coef group resp dpar nlpar   lb   ub
##           lkj(1)  cor
##           lkj(1)  cor              id
##           normal(0,0.5)    b setsize4          a <NA> <NA>
##           normal(0,0.5)    b setsize6          a <NA> <NA>
##           normal(0,0.5)    b setsize8          a <NA> <NA>
## student_t(3, 0, 2.5)    sd
## student_t(3, 0, 2.5)    sd              id          a    0
## student_t(3, 0, 2.5)    sd Intercept    id          a    0
## student_t(3, 0, 2.5)    sd setsize4    id          a    0
## student_t(3, 0, 2.5)    sd setsize6    id          a    0
## student_t(3, 0, 2.5)    sd setsize8    id          a    0
##           (flat)    b
##           normal(0,2)    b setsize4          c <NA> <NA>
##           normal(0,2)    b setsize6          c <NA> <NA>
##           normal(0,2)    b setsize8          c <NA> <NA>
## student_t(3, 0, 2.5)    sd
## student_t(3, 0, 2.5)    sd              id          c    0
## student_t(3, 0, 2.5)    sd Intercept    id          c    0
## student_t(3, 0, 2.5)    sd setsize4    id          c    0
## student_t(3, 0, 2.5)    sd setsize6    id          c    0
## student_t(3, 0, 2.5)    sd setsize8    id          c    0
##           normal(0,2)    b
##           normal(3,1)    b Intercept          c <NA> <NA>
##           normal(0,0.5)    b
##           normal(2,1)    b Intercept          a <NA> <NA>
##           constant(0)    b Intercept          b <NA> <NA>
##           source
##           default
## (vectorized)
## (vectorized)
## (vectorized)
## (vectorized)
##           default
## (vectorized)
## (vectorized)
## (vectorized)
## (vectorized)
## (vectorized)
##           default
## (vectorized)
## (vectorized)
## (vectorized)
##           default
## (vectorized)
## (vectorized)

```

```
## (vectorized)
## (vectorized)
## (vectorized)
##      user
##      user
##      user
##      user
##      user
```

Fit and run model

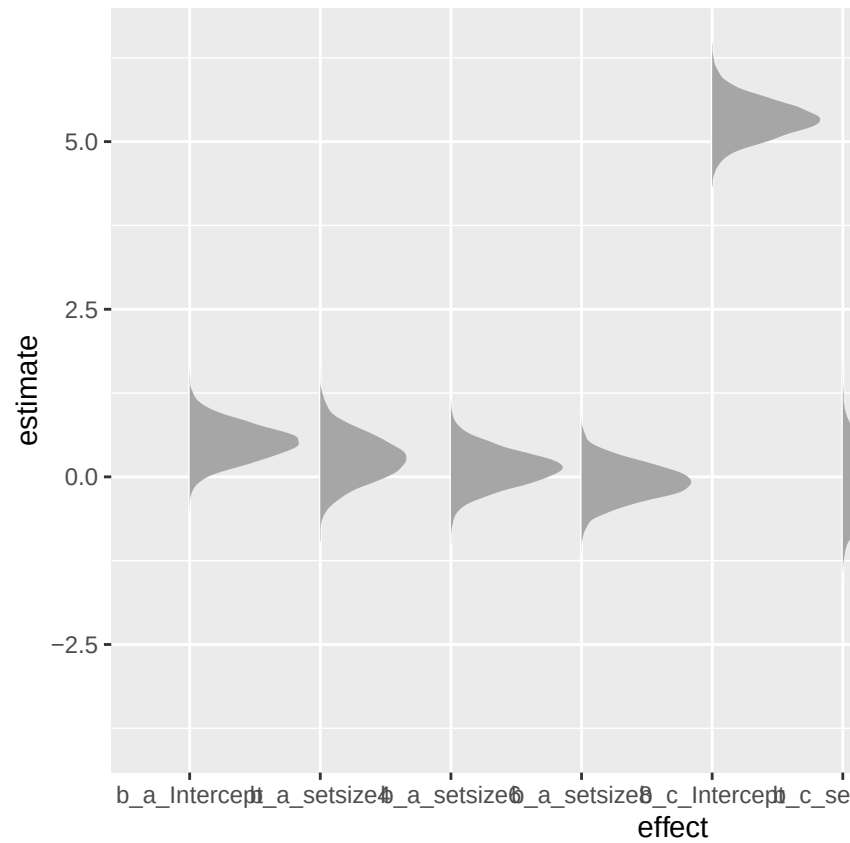
```
## Model: m3(resp_cats = c("IIP", "IOP", "NPL"),
##          num_options = c("n_IIP", "n_IOP", "n_NPL"),
##          choice_rule = "softmax",
##          version = "ss")
## Links: c = identity; a = identity
## Formula: b = 0
##          c ~ 1 + setsize + (1 + setsize | id)
##          a ~ 1 + setsize + (1 + setsize | id)
##          IIP ~ b + a + c
##          IOP ~ b + a
##          NPL ~ b
## Data: (Number of observations: 480)
## Draws: 4 chains, each with iter = 3000; warmup = 1000; thin = 1;
## total post-warmup draws = 8000
##
## Multilevel Hyperparameters:
## ~id (Number of levels: 20)
##
```

	Estimate	Est.Error	l-95% CI	u-95% CI	Rhat	Bulk_ESS
sd(c_Intercept)	0.80	0.19	0.47	1.22	1.00	2331
sd(c_setsize4)	0.94	0.39	0.27	1.80	1.00	1340
sd(c_setsize6)	0.40	0.21	0.06	0.90	1.01	533
sd(c_setsize8)	0.24	0.19	0.01	0.71	1.00	552
sd(a_Intercept)	0.31	0.14	0.07	0.62	1.00	2933
sd(a_setsize4)	0.49	0.39	0.02	1.43	1.00	3577
sd(a_setsize6)	0.25	0.18	0.01	0.67	1.00	2796
sd(a_setsize8)	0.16	0.13	0.01	0.48	1.00	2787
cor(c_Intercept,c_setsize4)	0.14	0.30	-0.43	0.75	1.00	2468
cor(c_Intercept,c_setsize6)	0.03	0.34	-0.61	0.70	1.00	1705
cor(c_setsize4,c_setsize6)	-0.12	0.35	-0.77	0.57	1.00	1892
cor(c_Intercept,c_setsize8)	-0.07	0.42	-0.77	0.76	1.00	3075
cor(c_setsize4,c_setsize8)	-0.03	0.43	-0.81	0.77	1.00	3221
cor(c_setsize6,c_setsize8)	0.41	0.45	-0.66	0.96	1.01	796
cor(a_Intercept,a_setsize4)	-0.00	0.43	-0.78	0.79	1.00	7863
cor(a_Intercept,a_setsize6)	-0.02	0.43	-0.79	0.78	1.00	6344
cor(a_setsize4,a_setsize6)	-0.03	0.44	-0.81	0.79	1.00	5260
cor(a_Intercept,a_setsize8)	-0.21	0.46	-0.90	0.72	1.00	6717
cor(a_setsize4,a_setsize8)	0.06	0.45	-0.79	0.83	1.00	5695
cor(a_setsize6,a_setsize8)	0.09	0.45	-0.79	0.85	1.00	5413
Tail_ESS						
sd(c_Intercept)	3090					
sd(c_setsize4)	1085					
sd(c_setsize6)	977					
sd(c_setsize8)	782					

```
## sd(a_Intercept)                2870
## sd(a_setsize4)                  3572
## sd(a_setsize6)                  4246
## sd(a_setsize8)                  4208
## cor(c_Intercept,c_setsize4)     2583
## cor(c_Intercept,c_setsize6)     2731
## cor(c_setsize4,c_setsize6)      3277
## cor(c_Intercept,c_setsize8)     4837
## cor(c_setsize4,c_setsize8)      4424
## cor(c_setsize6,c_setsize8)      1973
## cor(a_Intercept,a_setsize4)     5657
## cor(a_Intercept,a_setsize6)     5727
## cor(a_setsize4,a_setsize6)      5561
## cor(a_Intercept,a_setsize8)     5034
## cor(a_setsize4,a_setsize8)      5767
## cor(a_setsize6,a_setsize8)      6548
##
## Regression Coefficients:
##      Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
## c_Intercept      5.34      0.29    4.78    5.91 1.00    2829    3829
## c_setsize4      -0.11      0.39   -0.83    0.72 1.00    3128    4775
## c_setsize6      -2.17      0.27   -2.72   -1.65 1.00    3365    4078
## c_setsize8      -2.87      0.26   -3.40   -2.40 1.00    3391    4440
## a_Intercept      0.52      0.27   -0.00    1.05 1.00    3164    4442
## a_setsize4       0.28      0.34   -0.37    0.94 1.00    5926    5435
## a_setsize6       0.13      0.27   -0.41    0.66 1.00    3279    4632
## a_setsize8      -0.07      0.26   -0.58    0.43 1.00    3184    4484
##
## Constant Parameters:
##      Value
## b_Intercept    0.00
##
## Draws were sampled using sample(hmc). For each parameter, Bulk_ESS
## and Tail_ESS are effective sample size measures, and Rhat is the potential
## scale reduction factor on split chains (at convergence, Rhat = 1).
```

```
# extract posterior samples
posterior <- as_draws_df(m3_fit_ss_exp2) %>%
  select(starts_with("b_")) %>%
  pivot_longer(cols = starts_with("b_"), names_to = "effect", values_to = "estimate") %>%
  filter(effect != "b_b_Intercept")

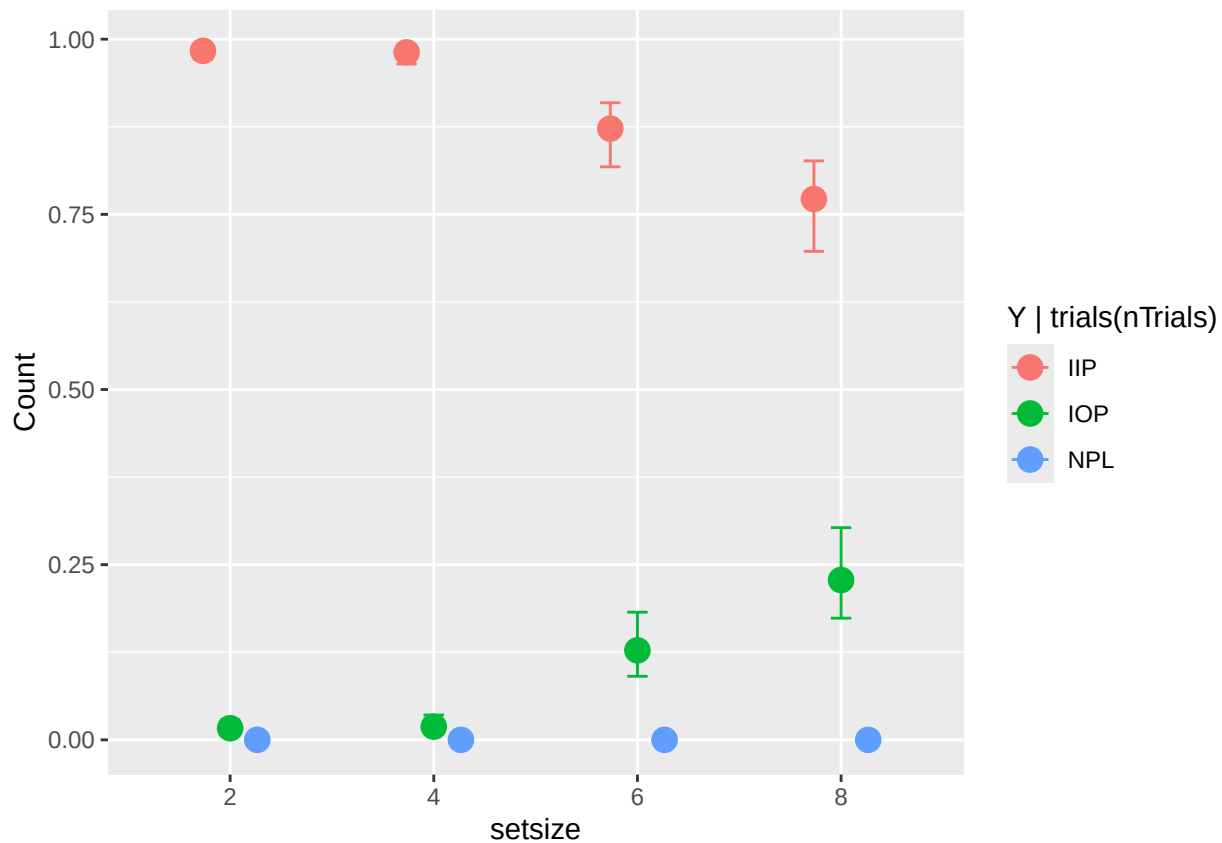
# plot posterior samples
ggplot(posterior, aes(x = effect, y = estimate)) +
  stat_slab()
```



Plot model posteriors and model predictions

plot model predictions on native scale

`conditional_effects(m3_fit_ss_exp2, categorical = T, effects = "setsize")`



```
# to-be testes hypotheses
tests <- c(
  "c_Intercept = c_setsize4",
  "c_setsize4 = c_setsize6",
  "c_setsize6 = c_setsize8",
  "a_Intercept = a_setsize4",
  "a_setsize4 = a_setsize6",
  "a_setsize6 = a_setsize8"
)

hypothesis(m3_fit_ss_exp2, tests)
```

Hypothesis tests

```
## Hypothesis Tests for class b:
##
```

	Hypothesis	Estimate	Est.Error	CI.Lower	CI.Upper	Evid.Ratio
## 1	(c_Intercept)-(c_... = 0	5.45	0.58	4.29	6.59	0.00
## 2	(c_setsize4)-(c_s... = 0	2.06	0.34	1.46	2.81	0.00
## 3	(c_setsize6)-(c_s... = 0	0.70	0.10	0.50	0.91	0.00
## 4	(a_Intercept)-(a_... = 0	0.24	0.52	-0.77	1.25	8.59
## 5	(a_setsize4)-(a_s... = 0	0.15	0.34	-0.48	0.84	2.05
## 6	(a_setsize6)-(a_s... = 0	0.20	0.15	-0.10	0.51	1.85

```
## Post.Prob Star
## 1      0.00    *
## 2      0.00    *
## 3      0.00    *
```

```
## 4      0.90
## 5      0.67
## 6      0.65
## ---
## 'CI': 90%-CI for one-sided and 95%-CI for two-sided hypotheses.
## '*': For one-sided hypotheses, the posterior probability exceeds 95%;
## for two-sided hypotheses, the value tested against lies outside the 95%-CI.
## Posterior probabilities of point hypotheses assume equal prior probabilities.
```